

Course 6041C

Consumer Electronics

Diploma Engineering Handbook / Notes

Program Elective - Semester 6 - Revision 2021

Audio Devices - Audio Systems - Video Systems - Home Appliances

Prepared for learning, revision, service understanding, and exam preparation.

INPUT

mic / camera

PROCESS

amplifier / MCU

OUTPUT

speaker / display

Polytechnic Study Hub

Course code kept exactly as 6041C. File: notes/downloadable-notes-6041C.pdf

What this subject covers

Consumer Electronics covers commonly used audio, video and home-appliance systems. A diploma student should understand the signal path, power path, main blocks, safety requirements, service symptoms, and the correct exam answer structure.

Learning outcomes

- Explain sound quantities, microphones, headphones and loudspeakers.
- Explain speaker enclosures, indoor acoustics, public address systems and digital audio.
- Explain television/display systems, cable/satellite TV basics and CCTV layout.
- Explain induction cooker, microwave oven, washing machine, air conditioner and refrigerator working principles.

How to study

- Study every equipment as INPUT - PROCESS - CONTROL - OUTPUT.
- Draw a labelled block diagram before writing long answers.
- Connect every topic to a practical service example.
- Use comparison tables for microphone, display and appliance questions.

Source note

This PDF is generated from the corrected standalone 6041C lesson source. It fixes the earlier issue where the 6041C page was an iframe wrapper and PDF extraction saw only a short text.

Sound fundamentals

Sound is a longitudinal mechanical wave. Important quantities are amplitude, wavelength, frequency, time period, velocity, intensity and decibel level. Human hearing is roughly 20 Hz to 20 kHz. Practical systems are judged by frequency response, distortion and noise.

Important formulae

- $v = f \times \lambda$, where v is wave velocity, f is frequency and λ is wavelength.
- $T = 1 / f$, where T is time period and f is frequency.
- $\text{dB} = 10 \log(P_2/P_1)$ for power ratio.

Microphones

A microphone converts acoustic energy into an electrical signal. Dynamic microphones use electromagnetic induction. Condenser and electret microphones use capacitance variation. Wireless microphones add RF transmitter and receiver sections.

Headphones and loudspeakers

Headphones and loudspeakers convert electrical audio signals into sound. A moving-coil loudspeaker uses a voice coil in a magnetic field. Multi-way systems use woofer, midrange, tweeter and crossover network.

Common exam points

- Draw microphone transducer diagram.
- Compare dynamic, condenser and electret microphones.
- Explain woofer, tweeter and crossover network.
- Write service faults: open coil, broken cable, jack fault, low output and distortion.

Speaker enclosures

A speaker cone radiates from the front and back. Rear wave cancellation reduces bass. Open baffle, sealed enclosure and bass reflex enclosure are used to improve low-frequency response and control sound radiation.

Indoor acoustics

Indoor sound quality depends on reflection, absorption, reverberation, echo and feedback. Auditoriums use acoustic panels, suitable speaker location, delay speakers and equalization to improve speech clarity.

Public address system

A PA system normally contains microphone, mixer, preamplifier, power amplifier, loudspeakers, monitor speakers and power supply. Important checks include impedance matching, cable routing, earthing and feedback control.

Digital audio

Digital audio uses sampling, quantization, encoding, storage and digital-to-analog conversion. Important terms are sampling frequency, bit depth, codec, compression and bitrate.

Service / field relevance

- Use balanced cables for long microphone lines when possible.
- Keep speaker cables away from high-current power wiring.
- Do not overload amplifier outputs with incorrect speaker impedance.
- Control feedback by microphone placement, speaker direction and gain setting.

Television block idea

A television receiver accepts RF, HDMI, AV, USB or network input. It processes the signal, separates audio and video, drives the display panel and provides speaker output. Modern TVs also include smart processor, Wi-Fi, memory and SMPS.

Display technologies

- LCD: liquid crystal shutters with a backlight.
- LED TV: LCD panel with LED backlight; LED strip failure can cause dark screen.
- OLED: self-emissive pixels with high contrast; burn-in risk exists.

Cable and satellite TV

Cable and satellite systems involve signal reception, demodulation, decoding, distribution and set-top box output. Service checks include input signal level, connector quality, LNB supply, cable continuity and HDMI output.

CCTV system

A CCTV installation uses camera, lens, image sensor, DVR or NVR, storage, monitor, network switch, power supply and cabling. IP CCTV also requires addressing, network configuration and storage management.

Expected diagrams

- TV receiver block diagram.
- LCD/LED/OLED comparison table.
- CCTV camera to DVR/NVR layout.
- Cable/satellite receiver basic block diagram.

Induction cooker

An induction cooker uses a high-frequency alternating magnetic field. Eddy currents and hysteresis losses heat the ferromagnetic vessel. Main blocks are rectifier, filter, inverter, resonant coil, microcontroller, temperature sensor and user interface.

Microwave oven

A microwave oven uses microwave energy to heat water molecules in food. Main blocks are high-voltage transformer or inverter, magnetron, waveguide, cavity, turntable motor, door interlock and controller. Door interlock safety is critical.

Washing machine

A washing machine contains motor drive, inlet valve, drain pump, drum/tub, pressure sensor, door lock, user interface and control board. Wash, rinse and spin operations are controlled by program logic.

Air conditioner and refrigerator

Both use vapour-compression cycle: compressor, condenser, expansion device and evaporator. Control parts include thermostat, sensors, fan motors, relay or inverter drive and protection devices.

Safety notes

- Isolate mains supply before service.
- Do not bypass door interlock or earth connection.
- Capacitors and inverter sections may retain dangerous charge.
- Use proper insulation and leakage checks before returning equipment to service.

Electrical power

$P = VI = I^2R = V^2/R$. Use for speaker load, heater load and appliance power estimation.

Energy consumption

Energy = Power x Time. For domestic billing, use kWh = watts x hours / 1000.

Sound wave relation

$v = f \times \lambda$. Used for frequency, wavelength and wave velocity relation.

Decibel rule

dB expresses a ratio on a logarithmic scale. For power ratio, $dB = 10 \log(P2/P1)$. For pressure ratio, $SPL = 20 \log(p/p0)$.

Service checklist

- Check supply, fuse, switch and visible damage first.
- Check connector, cable continuity and earthing before replacing boards.
- Check sensors and protection devices before suspecting the controller.
- Record symptoms, measurement values and final correction.

One and short answer questions

- Define frequency, wavelength, amplitude and decibel.
- What is the function of a crossover network?
- List the main parts of a PA system.
- What is reverberation?
- What is the role of DVR/NVR in CCTV?
- State two safety precautions for microwave oven servicing.

Descriptive questions

- Explain the working principle of a moving-coil microphone with diagram.
- Compare dynamic, condenser and electret microphones.
- Draw and explain the block diagram of a public address system.
- Compare LCD, LED and OLED display technologies.
- Draw and explain a CCTV system layout.
- Explain induction cooker working with block diagram.
- Explain vapour-compression cycle used in refrigerator and air conditioner.

Viva / practical style questions

- Why is speaker impedance matching important?
- Why does LED TV show sound but no picture when backlight fails?
- Why should microwave oven door interlock never be bypassed?
- How will you check a CCTV camera power fault?

Model paper pattern

Time: 3 Hours. Maximum Marks: 100. Part A: answer ten short questions. Part B: answer six medium questions. Part C: answer five descriptive questions with diagrams and applications.

Sample Part A

- Define dB and write one use.
- Name two microphone types.
- What is the function of woofer?
- State two blocks of a CCTV system.
- What is the use of thermostat?

Sample Part B

- Compare LCD and OLED display.
- Explain PA system block diagram.
- Explain induction cooker working principle.
- Write the main blocks of a washing machine.

Sample Part C

- Explain microphone types with comparison table and applications.
- Explain CCTV installation layout with service checks.
- Explain refrigerator cycle with block diagram and safety notes.

Answer writing method

For every long answer: definition, labelled block diagram, working explanation, application, common fault or safety note, and conclusion. For comparison answers, use a table with principle, supply, output quality, cost and application.

Final note

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